

JOONSUP KIM

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RESEARCH INTERESTS

Research Focus: Data-Driven Decision-Making; Machine Learning for Optimization; Market Design.

Methods and Applications: Online Learning and Bandit Algorithms; Matching Theory and Game Theory

EDUCATION

Seoul National University, Graduate School of Data Science

M.S. in Data Science; Advisor: Professor Min-hwan Oh

Current GPA: 4.02/4.30 (converted: 3.92/4.00)

Seoul, South Korea

March 2025 – February 2027 (expected)

Yonsei University

B.S. in Mathematics; B.A. in Economics (Double Major)

(Military Service: October 2022 – July 2024)

Seoul, South Korea

March 2019 – February 2025

- **Mathematics Department Rank:** 4th of 42
- **Overall GPA:** 3.95/4.30 (converted: 3.85/4.00)
- **Major GPA:** 4.03/4.30 (converted: 3.92/4.00)
Mathematics: 3.94/4.30 (converted: 3.89/4.00) *Economics:* 4.14/4.30 (converted: 3.95/4.00)

TEST SCORES

GMAT: 735 (Quantitative: 89; Verbal: 83; Data Insights: 88)

June 2026

TOEFL: 5.5 (Reading: 5.5; Listening: 6.0; Speaking: 5.0; Writing: 5.5)

May 2026

INTERNATIONAL BACKGROUND

Davis, California, United States: Lived in the U.S. for six years during childhood (grades 1–6).

RESEARCH EXPERIENCE

Seoul National University, Graduate School of Data Science

Research Assistant and M.S. Researcher; Advisor: Professor Min-hwan Oh

Seoul, South Korea

July 2024 – Present

- Conducting research on contextual matching under uncertainty, including problem formulation, algorithm development, and regret analysis.
- Studying bandit algorithms, online learning, reinforcement learning, and optimization for sequential decision-making.

Computational Science and Engineering Laboratory, Yonsei University

Undergraduate Research Assistant; Advisor: Professor Ji-hyun Lee

Seoul, South Korea

December 2021 – September 2022

- Analyzed theoretical epidemic models and studied how intervention and vaccination policies affect infection outcomes and economic costs.

RESEARCH IN PROGRESS

“Bilinear Matching Bandits with Sparse Edge Features and Sequential Feedback” (working title)

With Professor Min-hwan Oh. M.S. thesis; manuscript in preparation.

- Formulated a two-sided matching model motivated by ride-hailing, incorporating sparse feasible graphs, participant preference features, and pair-specific contexts through shared bilinear logistic acceptance models to enable learning across participants.
- Developed a UCB-based algorithm combining optimistic acceptance estimates with maximum-weight matching to maximize the expected number of mutually accepted pairs under sequentially censored feedback.
- Established a sublinear bound on expected cumulative regret relative to an oracle with known model parameters, characterizing its dependence on market size and feature dimensions.

COMPETITION AWARDS

Kakao AI Agent Competition – 2nd Place

December 2025

- Developed a persona-aware AutoReply Agent and a contextual-bandit Adaptive Push policy that learned push/no-push decisions from user reactions.

LG CNS Optimization Grand Challenge – Honorable Mention, Top 10%

July 2025

- Modeled a multi-vehicle loading, unloading, and repositioning problem as a graph and combined the Hungarian algorithm with evolutionary and local-search methods.

PROJECTS

Reinforcement Learning for Signal-Free Autonomous Driving

Seoul National University

Course Project: Reinforcement Learning

June 2026

- Co-developed a PPO-based controller for autonomous vehicles at a simulated four-way intersection without traffic signals, combining behavior cloning, curriculum learning, and a safety shield.
- Reduced mean delay by 38–80% and increased throughput by 2.8–3.4× versus fixed-time signals across three traffic-demand levels.

GPU-Accelerated Determinant Computation

Seoul National University

Course Project: Computing for Data Science II

December 2025

- Implemented GPU-parallel determinant computation using LU-based Gaussian elimination, progressively parallelizing row elimination, factor computation, pivot selection, row swaps, and diagonal-product reduction.
- Benchmarked six parallelization stages on matrices from $1,024 \times 1,024$ to $8,192 \times 8,192$, analyzing how memory access, synchronization, and kernel-launch overhead affect performance.

Semantic Search for Research Lab Recommendations

Seoul National University

Course Project: Big Data and Knowledge Management Systems I

June 2025

- Proposed the research-lab recommendation concept and curated faculty profiles, student affiliations, and publication records to support database construction.
- Collaborated on a GPT-based recommender combining SQLite and Pinecone with text embeddings and cosine similarity to recommend labs and papers.

TEACHING EXPERIENCE

Teaching Assistant

Seoul National University

Python for Data-Driven Business and Economics (English-language course)

July 2026

Teaching Assistant

Seoul National University

Probability and Statistics for Data Science

July 2025

Teaching Assistant

Seoul National University

Machine Learning and Deep Learning I (graduate course)

Fall 2025

AWARDS AND SCHOLARSHIPS

Academic Excellence Scholarship, Yonsei University

2024

Yonsei Honors Award, top 10% of college-year cohort

2024

Yonsei Honors Award, top 10% of college-year cohort

2022

Academic Excellence Scholarship, Yonsei University

2022

Mathematics Alumni Scholarship, Yonsei University

2021

Yonsei High Honors Award, top 3% of college-year cohort

2019

ADDITIONAL INFORMATION

Programming: Python, C, C++, SQL

Languages: Korean (native); English (fluent)

LEADERSHIP AND SERVICE

Military Service

October 2022 – July 2024

Academic Tutor in mathematics and science for underserved students

2021

Career Mentor for middle school students

2019

REFERENCES

Min-hwan Oh, Associate Professor

Graduate School of Data Science, Seoul National University

M.S. Advisor

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